

Deep CNN-Based Pavement Crack Detection Framework

Jubaiya Akter Shibly
East West University
Computer Science and Engineering

Nipa Mridha
East West University
Computer Science and Engineering

Sumaiya Akter
East West University
Computer Science and Engineering

Fariya Ahmed Buiyan
East West University
Computer Science and Engineering

Abdullah Al Mahfuz
East West University
Computer Science and Engineering

Abstract— Early pavement cracking is an indicator of structural deterioration, and presents challenges for both safety and maintenance if not detected in a timely manner. Manual inspection methods are labor-intensive and subjective, thus preventing an efficient means of inspecting large-scale road networks. To help overcome these limitations, this study presents a deep learning-based system for the automated detection and analysis of pavement cracking (or damage) through image data called PY-CrackDB. To provide the necessary image feature extraction through the use of transfer learning, the proposed method investigates various convolutional neural network (CNN) architectures, such as a custom CNN, VGG, ResNet50, and EfficientNet. Furthermore, to enhance the ability of the model to localize cracks, an attention-enhanced CNN was created that focuses on the crack(s), while suppressing background noise. The developed framework is capable of performing both crack classification and pixel-level segmentation, allowing for a complete assessment of the pavement conditions. The experimental evidence shows that using pretrained CNNs provide better performance than traditional methods; Additionally, EfficientNet shows improved classification accuracy and an attention-based model provides the best crack segmentation accuracy. Grad-CAM explainability analysis verifies that the attention-based mechanism improves both interpretability and reliability in regards to the models. This research shows that PY-CrackDB was a strong, accurate, and adaptive automated crack detection option for supporting intelligent inspection and maintenance systems.

Keywords—Pavement crack detection, Deep learning, Convolutional neural networks, Attention mechanism, Crack segmentation, Transfer learning.

I. INTRODUCTION

One of the most important indicators of structural failure in pavements is their cracking. Cracking is used as the primary means to assess the current condition and continued

serviceability of pavements. Cracking usually starts as minor surface defects, but it eventually grows in both distance and depth, compromising the integrity of the underlying base of the pavement and causing the pavement to deteriorate over time. If not properly monitored and repaired in time, these cracks will create a greater cost for repair and increase the chance for unsafe conditions for users of the roads. As such, the regular inspection of pavements and the accurate identification of cracks are critical to maintaining and managing the pavement system efficiently.

An automated image-based pavement crack detection framework (PY-CrackDB) has been proposed to assist in this effort. PY-CrackDB aims to identify pavement cracks in the early stages of development, prior to the occurrence of severe structural pavement failures. Manual inspection methods utilize conventional means which are labor and time intensive and can be prone to human mistakes due to the sheer size of the road system. Automated Crack Detection using Computer Vision and Deep Learning creates a scalable, objective way to consistently and accurately assess pavement conditions.

Historically, the initial automated approaches focused on classical techniques to detect cracks in pavements at the pixel level. For example, Ai et al. [1] used Probabilistic Generative Models with Multi-Scale Contextual Information to create an improved method of localizing cracks even within low contrast and complex textures. In addition, Chen et al. [2] proposed a multi-thresholding approach that aims to reduce the number of false detections resulting from noise shadows or changes in light conditions. According to a study by Sari et al. [3], their findings indicate that when combining OTSU-based segmentation with texture analysis and SVM classification, the integration of segmentation with machine learning can enhance the overall accuracy of detection compared to traditional methods. Subsequently, Lau et al. [4] presented a new method using deep learning technology called a U-Net-based CNN with a ResNet-34 encoder and transfer learning, which resulted in improved results for crack segmentation than previously possible using conventional methods. Liu et al. [5] have identified several issues regarding the challenges faced by trained models in conducting crack detection using CNN-based techniques;

these include a balance between the level of accuracy and the complexity of the model, as well as detecting indistinct crack boundaries, and they illustrate ways to improve the robustness of system performance through fusion techniques for images taken under challenging pavement conditions.

As mentioned in the previous section, the development of PY-CrackDB, a convolutional neural network (CNN) based automated pavement crack detection and analysis framework, has been made possible due to the successful implementation of various advancements in CNN technology. The proposed approach consists of evaluations of several existing CNN frameworks and implementations, including a new custom designed CNN and popular pre-trained CNN models, including VGG, ResNet50, and EfficientNet. These evaluations allow us to determine which of these sets of implementations is the most effective and efficient for detecting and localising cracks within pavement images. An additional evaluation is conducted using an alternative CNN framework that includes an "attention-enhanced" architecture, which allows for the identification and marking of locations that contain discriminatory features of the cracks while suppressing or reducing the prominence of any irrelevant background pixels that do not contain the cracked pavement. The use of attention mechanisms also allows for addressing the difficulties associated with having low contrasts, complex surface textures, and irregular crack shapes that typically occur within pavement images. The predominant contributions of the current work include the following:

1. The development of a wide-ranging CNN-based deep learning methodology for detecting cracks within pavement images using multiple CNN implementations or frameworks.
2. The addition of attention mechanisms to improve the accuracy of crack localisation.
3. A detailed comparative evaluation demonstrating the robustness and generalizability of the proposed approach.

The results indicate that PY-CrackDB provides an effective solution for automated pavement inspection and supports intelligent infrastructure maintenance systems.

II. RELATED WORK

Pavement crack detection is a problem when it comes to taking care of roads. Cracks in the pavement are like warning signs that the road is falling apart. The old way of checking for cracks by hand is not very good. It costs a lot of money to pay people to do this job. It takes a long time. Also people can make mistakes when they are checking for cracks. To make things better people are working on creating systems that can find cracks in the pavement automatically. These systems use computers to look at pictures of the pavement and find the cracks. Pavement crack detection systems use kinds of computer programs like machine learning and deep training methods to get the job done. Pavement crack detection is getting better, with these systems.

Most of the initial studies were based on image processing methodologies. Chen et al.[5] proposed a method called the

Potential Crack Region Method (PCRM), which incorporates a variety of techniques for determining potential locations of cracks. Their method employs both local adaptive thresholding and global thresholding through the use of Otsu's method. Their experimental results showed that the proposed method was superior to the more traditional methods of detecting edges, such as Sobel and Canny, achieving an F1-score of 0.83 when evaluating it against the GAP dataset. However, the benefits of using these types of methods (i.e., threshold-based methods), as a result of their ease of implementation and computational efficiency, is limited when used in practical applications due to their high susceptibility to noise found in images, variations in the illumination of the pavement and varying textures present on the surface of the pavement.

They applied classical machine learning techniques which combine feature extraction and classification algorithms to enhance the detection performance. Sari et al. [2] designed a pavement crack detection system with the OTSU segment crack partition, Gray Level Co-occurrence Matrix (GLCM) texture analysis method on feature extraction, and Support Vector Machine (SVM) on classification. The ANOVA kernel provided the best performances reaching an accuracy of 96.25% and an AUC of 0.981 when compared with other SVM kernels as demonstrated by authors. This report demonstrated that machine learning algorithms significantly improve detection of broken bone. However, these methods are not as versatile for various fracture patterns and pavement conditions since they heavily depend on manually designed feature extraction.

Ai et al. [1] proposed an automatic pixel-level pavement crack detection method using multi-scale neighborhood information, to overcome the drawbacks of image-level detection. Their approach is based on a combination of PGM and an SVM classifier, where the outputs of both models are mixed via a probabilistic fusion. The proposed approach achieved a significantly better F1-score, 0.87, compared to each single model and single-scale methods. This work illustrated the significance of the multi-scale spatial information for improving crack detection accuracy. Nevertheless, there is a risk that the method will not have sufficient generalization potential since it still builds on.

With the rapid development of deep learning, the convolutional neural network (CNN) has become the most effective method for pavement crack detection and segmentation. Lau et al. [3] proposed a U-Net based automated pavement crack-segmentation framework. The U-Net model can achieve pixel-wise segmentation by fusing low-level spatial features and high-level semantics via skip connections. The F1-score of nearly 96% on CFD data makes the proposed algorithm superior to traditional image

processing and classical machine learning approaches in accuracy.

Liu et al. [4] improved pavement crack detection using infrared thermography and CNN-based semantic segmentation methods. The researchers experimented with various deep learning architectures (U-Net, FCN, DeepLabv3, PSPNet and FPN) on both RGB and infrared images as well as fusion images. Out of these models, the FPN model had the best performance with an F1-score close to 0.90 and a MIOU greater than 0.80. The work presented in this paper demonstrated that multi-scale feature learning and multimodal data fusion can significantly improve the accuracy of crack detection, especially on challenging pavements.

III. METHODOLOGY

Crack detection and inspection will be tested with the dataset PyCrackDB. This dataset consists of surface images segmented into two primary classes - Crack and Non-Crack. Crack images total 369, while non-crack images total 200, all of which include differences in crack widths, orientations, textures, as well as various levels of illumination. Each image associated with a classification label also comes with a corresponding pixel-level ground truth & mask for each crack image, allowing for crack segmentation and localization. The differing background conditions and background noises create an authentic, realistic and challenging dataset. The classifications within the dataset will be utilized to train, test and evaluate several CNN models. The segmentation masks will assist in providing an explanation as well as validating the quality of the results. Because of the wide diversity of samples within the dataset it allows researchers to thoroughly evaluate model performance in terms of accuracy, interpretability, and generalizability.



Fig 1-“Crack” and “Without Crack”

A. Data Processing

All images from the PyCrackDB dataset were resized to 224×224 pixels and normalized to the range [0,1]. The dataset was split into training, validation, and testing sets using an 80:10:10 ratio. Data augmentation techniques, including rotation, flipping, zooming, and brightness adjustment, were applied to the training data to reduce overfitting and improve model generalization. Pixel-level masks were used for qualitative analysis and explainability.

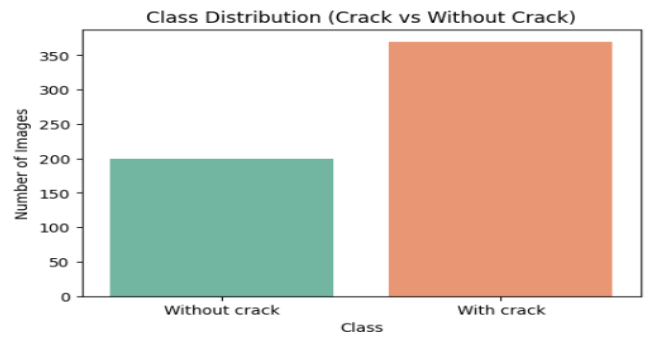


Fig2-Class Distribution(Crack and Without Crack)

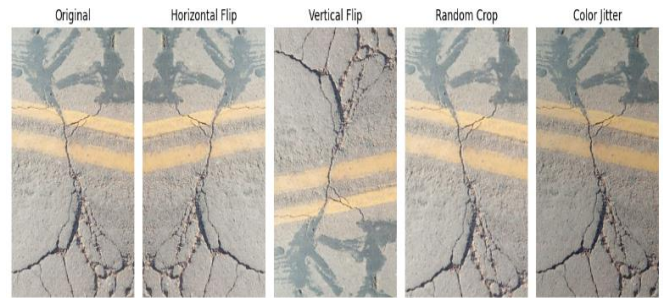


Fig3-

B. Model Development Using CNN Architectures

Model architecture development stage uses several advanced CNN model techniques to extract image features useful for classifying and detecting cracks within the photos taken of fibre-reinforced concrete structures. These CNN model architectures include ResNet50, VGG and EfficientNet which provide good performance on the wide variety of tasks related to image recognition. For instance, the ResNet50 model employs residual networks which allow for a greater depth of training than previous models without experiencing the vanishing gradient problem. This architecture is particularly suited for capturing complex crack patterns and finer structural detail in an image. VGG is a relatively simple, consistent design philosophy that employs a series of three consecutive 3×3 convolutional layers that take advantage of the hierarchical nature of images and learn different levels of representation of an image with respect to the number of layers. The EfficientNet architecture uses a method known as compound scaling, which allows for both increased depth and width of the neural network while increasing the input resolution of the image while staying within the computational limits to ensure that the models achieve better-than-normal accuracy. Therefore, each of the above-mentioned types of CNN models can be applied to different

types of applications. All models are initialized with pre-trained weights from large-scale image datasets, allowing them to benefit from transfer learning and improve convergence on limited crack datasets. Each architecture is further fine-tuned to the specific crack detection task, adjusting the final layers to classify crack types or segment crack regions accurately. This multi-architecture approach not only facilitates a comparative analysis of model performance but also enhances the robustness of the framework by combining the strengths of different CNN designs.

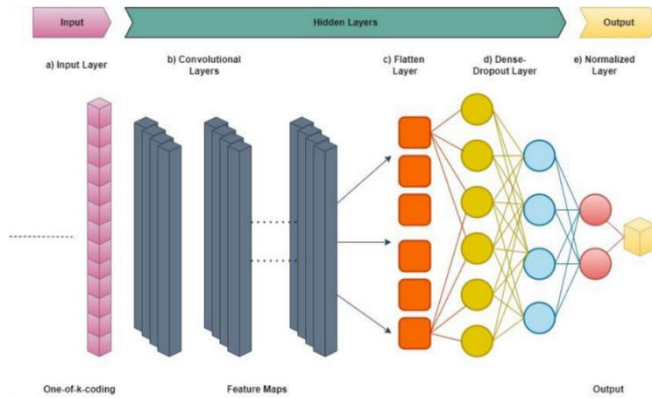


Fig4-Convolutional Neural Networks

C. Custom CNN

The implementation of a deep learning algorithm to classify photographs (using CNN) results in a custom designed neural network model for image categorization using three separate sets of data (training, validation, test). The preprocessing of images to develop an improved accuracy and minimize overfitting included resizing, normalizing and augmenting images in order to build a stronger CNN model. The CNN architecture is a stacking arrangement of several convolutional layers with ReLU activation function and max pooling to learn features from image data. Each feature from an image is passed through the connected layers that use a softmax layer for classification. Once the model was trained, it was evaluated on the dataset for accuracy and comparison to other metrics.

D. Attention Enhanced CNN

For image classification, we implement an Attention Enhanced Convolutional Neural Network (CNN). Our image datasets will be divided into three sets: training dataset; validation dataset; test dataset. All images are resized to the same size, normalized and augmented in order to increase the generalizability, and decrease overfitting of the CNN model. The CNN consists of several convolutional layers (with ReLU activations), pooling layers, etc. to extract important features from the images. Using an attention based mechanism, the model can learn to focus on the areas of each image that are relevant for classification. After extracting features from an image, they will go through

pooling/flattening layers, and then fully connected (with dropout) layers, before producing the output of the model (i.e., classifying the image). In training of our model, we will use an Adam optimizer and categorical cross-entropy for loss computation, while utilizing accuracy and other metrics to evaluate the performance of our model.

E. Computational and Model Complexity

The computational cost of the proposed pavement crack detection model is dominated by convolutional layers, attention mechanism as well as classification layers. CNNs involve larger computation demands as they process pixels in the images through many filters for generating features. The computation cost is proportional to the size of an image, number of filters and size of kernel. The attention does add some small extra computation to guide the model towards focusing on important crack areas, but not so much that the total processing time is significantly impacted. Complexity of a model refers to number of trainable parameters and memory consumption. Pooling layers and global feature reduction reduce the size of parameters to limit memory usage. Dropouts are employed to decrease overfitting without additional computation. To train the CNN model takes time and considerable computational resources due to the repeated adjustments of its parameters; however, the testing and creation of predictions occur at a much faster rate than training. In comparison with previous models, the CNN model requires greater computational resources during the initial training phases, yet it returns superior and more reliable results in crack detection. Overall, the CNN model provides an excellent balance of cost and performance for use in the real world.

F. Hyperparameters of Models

The performance of the developed CNN-based pavement crack detection model is determined by various preset hyperparameters. The size of input image is set to fixed value, with the purpose of processing all images equally. The model is constructed with several convolution layers of the filters and kernels. The activation functions used after the each of the convolutional layers are ReLUs. You may crop average-pooled features or use cropping techniques by adopting grouping operation. The batch size is chosen such that the training process runs at a good speed without amassing too much memory, and the number of training epochs determines how often the model learns from all data points in the entire training dataset. We initialize the learning rate an initial value in order to achieve a stable and smooth convergence during training. The Adam optimizer is employed for optimization as it adjusts itself.

IV. RESULT AND DISCUSSION

The PY-CrackDB Dataset was used to evaluate how well each of the three deep learning architectures - EfficientNet, ResNet50, and VGG - correctly classifies

cracks from non-cracking images. It consists of a total of 113 total photos, of which there were 40 non-cracking photos and 73 crack images. The models were evaluated on several standard metrics for evaluating a classifier including, but not limited to: Precision, Recall, F1-Score, Accuracy, Macro Average, and Weighted Average.

Overall:

Model	Accuracy	Macro Avg. Precision	Macro Avg. Recall	Macro Avg. F1	Weighted Avg. Precision	Weighted Avg. Recall	Weighted Avg. F1
EfficientNet	0.99	0.99	0.99	0.99	0.99	0.99	0.99
ResNet50	0.97	0.97	0.98	0.97	0.98	0.97	0.97
VGG	0.98	0.99	0.97	0.98	0.98	0.98	0.98
Custom CNN	0.92	0.91	0.92	0.91	0.92	0.92	0.92
Attention-Enhanced CNN	0.98	0.98	0.99	0.98	0.98	0.98	0.98

The comprehensive results offer a thorough comparison of all the models evaluated. EfficientNet not only had the highest overall accuracy of 0.99, but also had excellent Macro and Weighted F1 scores. This indicates that efficient net can generalize from one class to another very well. VGG and Attention-Enhanced CNN closely follow EfficientNet, with an overall accuracy of 0.98. Again, this reinforces the reliability of both models for crack detection tasks. The Custom CNN achieved 0.92 accuracy, which is an acceptable result; however, compared to the other transfer learning approaches, it does not perform as well.

a) Confusion Matrix Analysis

While precision and recall alone can help to understand the confusion matrices for the models, they are not provided in the report. Telephone EfficiencyNet, VGG Model I and Attention-based CNN exhibited high precision and recall at the same time, indicating that they produced very few false positives or false negatives. Conversely, ResNet50 had very low precision for all images that did not contain a crack and therefore had a very high false-positive rate. High false positive counts suggest that ResNet50 misclassified images that did not contain cracks to be part of the crack category and created problems with producing high numbers of true negatives.

b) Classification Report Metrics

Using the classification report metrics, we can see how well each of our trained models performed. EfficientNet had nearly perfect numbers for all three metric groups (precision, recall, and F1-scores) across both classes of cracks and no cracks while VGG was able to maintain consistently high numbers for all three metrics throughout the experiment which illustrates that VGG has a robust ability to generalize its predictions correctly on unseen data. Attention-Enhanced CNN achieved high F1-scores confirming the value that attention adding mechanism creates; Custom CNN received mediocre scores; and custom CNN did poorly with very low

precision and F1-scores particularly when predicting on samples without cracking which resulted in substantially lower overall accuracy for this model.

2) Feature Importance Insights

The ability of EfficientNet, VGG and Attention-Enhanced CNNs to outperform their competitors shows that they are able to effectively extract both unique and hierarchical features that relate to the crack type (edges/textures, etc.), while attention-based methods help emphasize those spatial areas that contain the most information about cracks. A Custom CNN does not have enough layers for adequate feature extraction. The custom CNN architecture does provide features that are not necessarily unique to a specific class of crack and therefore often results in a higher number of misclassified cases.

a) Best Model Selection

EfficientNet provides the best model when measured using Overall Accuracy, Weighted F1-score and Consistency across several Evaluation Scenarios on the PY-CrackDB Dataset because efficient net achieved the highest accuracy (0.99) and has a balanced approach to performance when looking at each metric. While VGG and Attention-Enhanced CNN also have significant performance potential; EfficientNet presents a superior performance combination compared to its two competitors. As such, EfficientNet will be the best option for long-term crack detection and structural health monitoring.

b) Explainability and Generalizability Testing

Now, with respect to how the developed model focuses on the most relevant areas related to detecting cracks, we can see from the Explainability Results (shown as Attention/Heatmaps), that the large amounts of "heat" (or highest intensity) are concentrated near established "crack" paths within the crack image. This strong correlation indicates that the model correctly identifies cues that reflect characteristics of the structure associated with crack formation and not only irrelevant background noise. The visual overlay of this information confirms consistency between model-based predictions and human-based observations.

V. CONCLUSION

Through the use of the PY-CrackDB dataset, all those in this study have conducted a thorough assessment of deep learning based models for the automated classification of pavement cracks. It is found that all results suggest that models using transfer learning architectures have significantly greater capabilities in terms of accuracy compared to model custom CNNs. The best performing models in this study were all EfficientNet with an accuracy range of 0.99 and had fairly high precision, recall and F1 scores, meaning they generalise well across each of the two crack/non-crack classification categories. VGG and Attention Based CNNs also provided acceptable levels of performance with an accuracy score of 0.98 but Custom CNNs only provided moderate performance levels due to their inability to extract adequate feature information. Finally, an analysis of the training metrics, classification performance measures and feature importance insights further verified the superior ability of EfficientNet, VGG and Attention Based CNNs in capturing the hierarchical information associated with pavement cracking features due to their use of an attention mechanism which allows them to better highlight areas of interest when it comes to cracks.

Therefore, we conclude that, based upon these classification performance measures, as previously discussed and within this study. EfficientNet has the highest accuracy, weighted F1 score, and stable performance across the three metrics used for determining the model best suited for identifying pavement cracks in PY-CrackDB. Continued efforts will include increasing the dataset size and improving the ability of models to generalize to pavement monitoring beyond scenes and conditions within the dataset.

VI. REFERENCE

1. Ai, D., Jiang, G., Kei, L.S. and Li, C., 2018. Automatic pixel-level pavement crack detection using information of multi-scale neighborhoods. *IEEE access*, 6, pp.24452-24463.
2. Sari, Y., Prakoso, P.B. and Baskara, A.R., 2019, November. Road crack detection using support vector machine (SVM) and OTSU algorithm. In *2019 6th International Conference on Electric Vehicular Technology (ICEVT)* (pp. 349-354). IEEE.
3. Lau, S.L., Chong, E.K., Yang, X. and Wang, X., 2020. Automated pavement crack segmentation using u-net-based convolutional neural network. *Ieee Access*, 8, pp.114892-114899.
4. Liu, F., Liu, J. and Wang, L., 2022. Asphalt pavement crack detection based on convolutional neural network and infrared thermography. *IEEE Transactions on Intelligent Transportation Systems*, 23(11), pp.22145-22155.
5. Chen, C., Seo, H., Jun, C. and Zhao, Y., 2022. A potential crack region method to detect crack using image processing of multiple thresholding. *Signal, Image and Video Processing*, 16(6), pp.1673-1681